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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 20

Construct a C program to simulate Reader-Writer problem using semaphores

PROGRAM:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

sem_t mutex, writeBlock;
int data = 0, readersCount = 0;

void *reader(void *arg)
{
    for (int i = 0; i < 10; i++)
    {
        sem_wait(&mutex);
        readersCount++;

        if (readersCount == 1)
            sem_wait(&writeBlock);

        sem_post(&mutex);

        printf("Reader reads data: %d\n", data);

        sem_wait(&mutex);
        readersCount--;

        if (readersCount == 0)
            sem_post(&writeBlock);
```

```
        sem_post(&mutex);
    }
    return NULL;
}

void *writer(void *arg)
{
    for (int i = 0; i < 10; i++)
    {
        sem_wait(&writeBlock);

        data++;
        printf("Writer writes data: %d\n", data);

        sem_post(&writeBlock);
    }
    return NULL;
}

int main()
{
    pthread_t r, w;

    sem_init(&mutex, 0, 1);
    sem_init(&writeBlock, 0, 1);

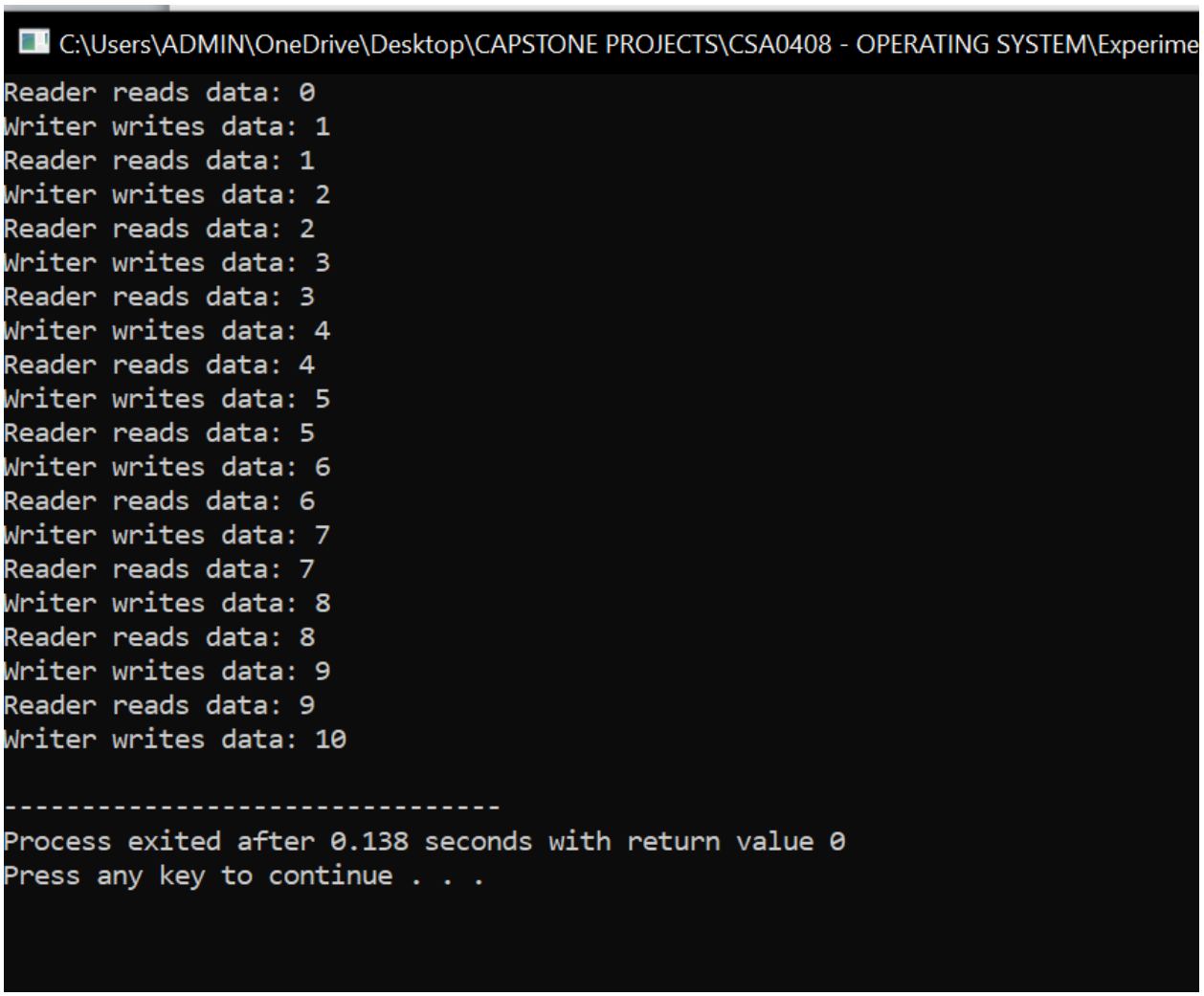
    pthread_create(&r, NULL, reader, NULL);
    pthread_create(&w, NULL, writer, NULL);

    pthread_join(r, NULL);
    pthread_join(w, NULL);

    sem_destroy(&mutex);
    sem_destroy(&writeBlock);

    return 0;
}
```

OUTPUT:


C:\Users\ADMIN\OneDrive\Desktop\CAPSTONE PROJECTS\CSA0408 - OPERATING SYSTEM\Experime
Reader reads data: 0
Writer writes data: 1
Reader reads data: 1
Writer writes data: 2
Reader reads data: 2
Writer writes data: 3
Reader reads data: 3
Writer writes data: 4
Reader reads data: 4
Writer writes data: 5
Reader reads data: 5
Writer writes data: 6
Reader reads data: 6
Writer writes data: 7
Reader reads data: 7
Writer writes data: 8
Reader reads data: 8
Writer writes data: 9
Reader reads data: 9
Writer writes data: 10

Process exited after 0.138 seconds with return value 0
Press any key to continue . . .